

A REFERENCED STATUTE ARTICLES

Civil Act Article 750 (Definition of Torts)

Any person who causes losses to or inflicts injuries on another person by an unlawful act, intentionally or negligently, shall be bound to make compensation for damages arising therefrom.

Civil Act Article 766 (Prescription in respect of Right to Claim for Damages)

(1) The right to claim for damages resulting from an unlawful act shall lapse by prescription if not exercised within three years commencing from the date on which the injured party or his/her legal representative becomes aware of such damage and of the identity of the person who caused it.

(2) The provisions of paragraph (1) shall also apply if ten years have elapsed from the time when the unlawful act was committed.

(3) Where a minor suffers from sexual violence, sexual molestation, sexual harassment, other sexual infringement, the extinctive prescription of the right to claim compensation for damages shall not proceed until such minor attain the majority.

B SYSTEM PROMPTS

This appendix provides the exact prompts used in each experimental condition. All models received identical prompt structures, with only the model-specific API parameters differing. English translations are provided below; original Korean prompts are available in the source repository.

B.1 Closed Book

In the Closed Book condition, models received only the question and answer format instruction without any external context or tool access.

```
{question}

**Important: You must provide your final answer in the following
format:**
answer: [a number from 1-5]

Example:
answer: 3
```

B.2 Naïve RAG

In the Naïve RAG condition, retrieved context (10 case summaries and 10 statute articles) was prepended to the question.

```
{retrieved_context}

---

{question}

**Important: You must provide your final answer in the following
format:**
answer: [a number from 1-5]

Example: answer: 3
```

The retrieved_context includes statute articles and case summaries formatted with metadata (case number, court name, decision date, judgment summary, reference statutes, citation count, and relevance score).

B.3 Agentic RAG (Without Guidance)

In the main agentic RAG experiments, models received minimal instruction about MCP tool availability. This prompt was used for all results reported in Table 2.

Solve the following bar exam question using MCP to search for relevant case law and statutes.

{question}

****Important: You must provide your final answer in the following format:****

answer: [a number from 1-5]

Example:

answer: 3

B.4 Agentic RAG with Prompt Guidance

For the prompt guidance experiment (Section 5.4), Gemini 2.5 Pro received explicit step-by-step guidance on tool usage.

You are a Korean legal expert. To solve the legal question below, you must use MCP tools to search and analyze relevant statutes and case law.

Problem-Solving Procedure

1. Analyze the question to identify relevant statutes and issues
2. search_statutes to find relevant laws get_statute_content to review articles
3. search_cases to find relevant precedents get_case_content to review case details
4. Repeat the above process with additional searches as needed
5. Derive final answer after securing sufficient legal grounds

* If you don't know the structure of a statute or specific article numbers, first use list_statute_articles to check the table of contents.

****Important: Call tools repeatedly for accurate answers. At least 10 tool calls are recommended.****

Question

{question}

****You must provide your final answer in the following format:****

answer: [a number from 1-5]

Example: answer: 3

C MCP TOOL SPECIFICATIONS

The MCP server provides five tools designed to mirror legal experts’ research workflow. Tools are categorized into *Search* tools (returning ranked lists) and *Lookup* tools (returning full content). English translations of tool responses are provided below; original Korean outputs are available in the source repository.

C.1 Tool Definitions

Table 6 summarizes all five tools. Search tools use hybrid retrieval combining BM25 and semantic similarity with Reciprocal Rank Fusion (RRF).

Table 6: MCP tool definitions. Required parameters marked with *.

Tool	Parameters
<i>Search Tools</i>	
search_cases	query*: string (keywords) reference_statute: string court_name: string date_from, date_to: YYYYMMDD
search_statutes	query*: string (keywords) law_type: string
<i>Lookup Tools</i>	
get_case_content	case_number*: string
get_statute_content	statute_id*: string article_number: string article_range: string (e.g., “1-10”)
list_statute_articles	statute_id*: string

C.2 Quick Access Statute IDs

Table 7 provides statute IDs for major Korean laws, enabling direct lookup without prior search.

Table 7: Quick access statute IDs for major Korean laws.

Statute	ID	Statute	ID
Constitution	1444	Framework Act on Administration	14041
Civil Act	1706	Administrative Procedure Act	1362
Commercial Act	1702	Administrative Litigation Act	1363
Civil Procedure Act	1700	Constitutional Court Act	11233
Criminal Act	1692		
Criminal Procedure Act	1671		

C.3 Example Tool Calls and Responses

search_cases

```
// Call
{"name": "search_cases", "args": {"query": "defamation damages",
  "court_name": "Supreme Court"}}

// Response (abbreviated)
```

```
Available judgments (top 5 matches):
-----
- case_number: 2020Da12345
- case_name: Damages
- court_name: Supreme Court
- decision_date: 2021-03-15
- judgment_summary: In cases of defamation through internet
  postings...
- reference_statutes: Civil Act Art. 750, Art. 751
- citation_count: 127
- token_count: 12,453
-----
```

get_case_content

```
// Call
{"name": "get_case_content", "args": {"case_number":
  "2020Da12345"}}

// Response (abbreviated)
=====
CASE: 2020Da12345
=====
Title: Damages
Court: Supreme Court
Date: 2021-03-15
Tokens: 12,453

=====
JUDGMENT TEXT
=====
[Holding]
When defamatory content is posted on an internet bulletin board...
[Reasoning]
For liability for defamation damages to be established...
[full judgment text continues...]
```

search_statutes

```
// Call
{"name": "search_statutes", "args": {"query": "personal information
  protection"}}

// Response (abbreviated)
Available statutes (top 5 matches):
-----
- statute_id: 20547
- law_name: Personal Information Protection Act
- abbreviation: PIPA
- law_type: Act
- clause_count: 75
- description: This Act aims to protect the freedom and rights of
  individuals...
- citation_count: 1,234
- token_count: 45,621
-----
```

get_statute_content

```
// Call
{"name": "get_statute_content", "args": {"statute_id": "1706", "article_number": "750"}}

// Response
=====
STATUTE: 1706
=====
Law Name: Civil Act
Law Type: Act
Retrieved: Article 750

=====
STATUTE TEXT
=====
Article 750 (Torts) [Citations: 8,234]
Any person who causes damage to another by an unlawful act,
intentionally or negligently, shall be liable to compensate for such
damage.
```

list_statute_articles

```
// Call
{"name": "list_statute_articles", "args": {"statute_id": "1706"}}

// Response (abbreviated)
=====
STATUTE ARTICLES LIST: Civil Act
=====
Statute ID: 1706
Total Articles: 1,118

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TABLE OF CONTENTS
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Article 1 (Sources of Law) [Citations: 342]
Article 2 (Good Faith) [Citations: 1,523]
...
Article 750 (Torts) [Citations: 8,234]
Article 751 (Non-Pecuniary Damages) [Citations: 2,156]
...
```

D DATABASE AND RETRIEVAL CONFIGURATION

D.1 Data Sources and Statistics

Table 8 summarizes the legal database used in this study, collected from the Korea Law Information Center (<https://open.law.go.kr>).

Table 8: Legal database statistics.

Data Type	Count
Court Cases	163,546
Constitutional Court Decisions	29,730
Statutes	5,474
Statute Articles	200,633
Total Cases	193,276

D.2 Elasticsearch Index Configuration

The system uses Elasticsearch 8.19.5 with three indices. Text fields use the Nori analyzer for Korean morphological analysis. Table 9 shows key fields for each index.

Table 9: Key fields in Elasticsearch indices.

Index	Field	Type
Court Cases	case_number	keyword
	case_name	text (nori)
	court_name	keyword
	decision_date	date (yyyyMMdd)
	judgment_summary	text (nori)
	embedding_vector	dense_vector (4096d)
Statutes	statute_id	keyword
	law_name	text (nori) + keyword
	clause_number	keyword
	clause_content	text (nori)
	embedding_vector	dense_vector (4096d)

D.3 Embedding Model

Text embeddings are generated using the Upstage embedding-query model, producing 4,096-dimensional vectors. Vector indices use HNSW (Hierarchical Navigable Small World) with cosine similarity.

D.4 Hybrid Search Parameters

Search tools use Reciprocal Rank Fusion (RRF) to combine BM25 lexical search and semantic vector search. Table 10 details the configuration.

Table 10: Hybrid search parameters.

Component	Parameter	Value
RRF Fusion	k (ranking constant)	60
	BM25 weight	1.05
	Vector weight	1.0
KNN Search	k (neighbors)	30
	num_candidates	150
Results	Cases returned	5
	Statutes returned	10

D.5 BM25 Field Boosting

Table 11 shows field-level boosting weights for BM25 search.

Table 11: BM25 field boosting weights.

Case Search		Statute Search	
Field	Boost	Field	Boost
judgment_summary	6.0	law_name	3.0
reference_statute	3.5	abbreviation	2.0
case_content	2.5	description	1.0
case_number	1.5		
case_name	0.8		

E MODEL API PARAMETERS

Table 12 summarizes the API parameters used for each model configuration. All experiments were conducted in December 2025.

Table 12: Model API parameters.

Model	Config	Temp	Reasoning
GPT-5.1	None	0	effort: none
	Low	1	effort: low
	Medium	1	effort: medium
	High	1	effort: high
Claude 4.5	Non-Think	0	(disabled)
	Max-Think	1	budget: 63,999
Gemini 2.5	Low	1	budget: 128
	High	1	budget: 32,768

Notes.

- **GPT-5.1:** OpenAI Responses API. max_output_tokens=64,000.
- **Claude Sonnet 4.5:** Version claude-sonnet-4-5-20250929. max_tokens=64,000. Extended Thinking requires a temperature of 1.
- **Gemini 2.5 Pro:** MCP via automatic_function_calling with maximum_remote_calls=99.

F STATISTICAL SIGNIFICANCE TESTS

To verify that observed performance differences between Naïve RAG and agentic RAG are not due to chance, we conducted McNemar’s test and bootstrap resampling.

McNemar’s Test. McNemar’s test evaluates whether the disagreement pattern between two systems is symmetric. For paired binary outcomes—correct or incorrect responses on the same 150 questions—it tests whether the number of questions where only agentic RAG succeeded ($N \rightarrow A$) differs significantly from those where only Naïve RAG succeeded ($A \rightarrow N$).

Bootstrap Confidence Intervals. We computed 95% confidence intervals for each system’s accuracy using bootstrap resampling with $n = 10,000$ iterations.

Table 13: Statistical significance of performance differences. McNemar’s test evaluates disagreement asymmetry; Bootstrap ($n = 10,000$) provides 95% CI. Bold indicates the superior method.

Model	McNemar’s Test				Accuracy [95% CI]	
	$N \rightarrow A$	$A \rightarrow N$	χ^2	p	Naïve	Agentic
Claude 4.5 (Max)	18	2	11.25	<.001	84.0 [78.0, 89.3]	94.7 [90.7, 98.0]
GPT-5.1 (High)	19	3	10.23	.001	86.0 [80.0, 91.3]	96.7 [93.3, 99.3]
Gemini 2.5 (High)	4	28	16.53	<.001	89.3 [84.7, 94.0]	73.3 [66.0, 80.0]

$N \rightarrow A$: Naïve incorrect $\rightarrow$ Agentic correct; $A \rightarrow N$: Agentic incorrect $\rightarrow$ Naïve correct.

Results. All three comparisons showed statistically significant differences ($p < .01$). For Claude and GPT, agentic RAG significantly outperformed Naïve RAG, with non-overlapping 95% confidence intervals. For Gemini, the pattern reversed: Naïve RAG significantly

outperformed agentic RAG, confirming that the observed performance degradation is not due to sampling variability.

G SAMPLE QUESTION WITH TOOL TRACES

This appendix presents a sample question (Q10) demonstrating the three distinct tool-use patterns observed across model families. The original Korean question and tool-call logs are shown below to ensure reproducibility.

G.1 Question

문 10. 「부동산 실권리자명의 등기에 관한 법률」이 적용되는 명의신탁에 관한 설명 중 옳은 것은? (다툼이 있는 경우 판례에 의함)

① 3자 간 등기명의신탁에서 명의수탁자가 명의신탁된 부동산을 임의처분하여 제3자가 그 소유권을 취득한 경우, 매도인의 소유권이전등기의무가 이행불능이 되어 발생하는 매도인과 명의신탁자 사이의 법률관계와 명의수탁자가 매도인의 소유권을 침해하여 발생하는 명의수탁자와 매도인 사이의 법률관계를 각각 구분하여 개별적으로 이해관계를 조정하면 부당이득반환 제도의 취지에 배치될 수 있다.

② 3자 간 등기명의신탁에서 명의신탁자가 매도인을 대위하지 않고 직접 명의수탁자를 상대로 부당이득반환을 원인으로 한 소유권이전등기를 청구한 경우, 이에 따라 마쳐진 명의신탁자 명의 소유권이전등기는 무효이다.

③ 계약명의신탁에서 매도인이 명의신탁약정에 대하여 알지 못했던 경우, 명의수탁자가 명의신탁자에 대한 매수자금 반환에 갈음하여 명의신탁된 부동산 자체를 양도하기로 합의하고 그에 기하여 명의신탁자가 지정하는 제3자 앞으로 소유권이전등기를 마쳐 주었다면 그 제3자 명의 소유권이전등기는 유효이다.

④ 계약명의신탁에서 매도인이 명의수탁자와 매매계약을 체결할 때는 명의신탁약정에 대하여 알지 못하였으나 명의수탁자 명의로 소유권이전등기를 마쳐 줄 때는 이를 알게 된 경우, 매도인과 명의수탁자 간 매매계약은 소급적으로 무효가 된다.

⑤ 계약명의신탁에서 매도인이 명의신탁약정에 대하여 알고 있었던 경우, 매도인과 명의수탁자가 체결한 매매계약은 원시적으로 무효이고 해당 부동산의 소유권은 매도인에게 그대로 남아 있게 되므로 특별한 사정이 없는 한 명의신탁자는 매도인에게 소유권이전등기를 청구할 수 있다.

G.2 Tool-Call Comparison

Table 14 summarizes each model’s tool usage on Q10.

Table 14: Tool-call comparison for Q10.

Model	Total	search	get_case	Lookup	Correct
GPT-5.1	27	15	10	66.7%	✓
Claude 4.5	8	4	2	50.0%	✓
Gemini 2.5	5	5	0	0.0%	×

G.3 Tool-Call Sequences

GPT-5.1 (High) – 27 calls, Correct (①)

1. search_cases("3자 간 등기명의신탁 임의처분 부당이득반환 법률관계 구분")
2. search_cases("3자간 등기명의신탁 임의처분 부당이득반환 제도의 취지")

3. get_case_content("2018다284233")
4. get_case_content("2018다284233")
5. get_case_content("2018다284233")
6. search_cases("각각 구분하여 개별적으로 이해관계를 조정하면 부당이득반환 제도의 취지에 배치")
7. get_case_content("2008다55290,55306")
8. search_cases("3자간 등기명의신탁 명의신탁자가 명의수탁자를 상대로 부당이득을 원인으로 한 소유권이전등기 무효")
9. get_case_content("2004다6764")
10. search_cases("명의신탁자가 매도인을 대위하지 않고 명의수탁자를 상대로 부당이득반환을 원인으로 한 소유권이전등기청구 등기 무효")
11. get_case_content("2006다73102")
12. search_cases("3자간 등기명의신탁 명의수탁자가 명의신탁자 앞으로 소유권이전등기를 경료해 준 경우 실체관계에 부합하는 등기 유효")
13. search_cases("부당이득반환을 원인으로 한 소유권이전등기 3자간 등기명의신탁")
14. get_case_content("2011다103472")
15. get_case_content("2014다30483")
16. search_cases("계약명의신탁 매도인이 명의신탁약정을 알지 못 명의수탁자가 매수자금 반환에 갈음하여 부동산을 양도 제3자 명의 소유권이전등기 유효")
17. search_cases("계약명의신탁 대물급부 제3자 앞으로 소유권이전등기 유효")
18. search_statutes("부동산 실권리자명의 등기에 관한 법률")
19. get_statute_content("1178", "4")
20. search_cases("계약명의신탁 매도인이 매매계약 체결시에는 명의신탁약정을 알지 못하였으나 소유권이전등기 시에는 알게 된 경우")
21. get_case_content("2017다257715")
22. get_case_content("2010다95185")
23. search_cases("매도인의 소유권이전등기의무가 이행불능이 되어 발생하는 매도인과 명의신탁자 사이의 법률관계")
24. search_cases("3자간 등기명의신탁 명의신탁자 명의로 소유권이전등기 실체관계에 부합하는 유효한 등기인지 여부")
25. search_cases("매도인이 명의신탁약정에 대하여 알지 못 매수자금 반환에 갈음하여")
26. search_cases("대물급부 제3자 앞으로 소유권이전등기 명의신탁")
27. search_cases("계약명의신탁 제3자 명의 소유권이전등기 유효 여부")

1. search_cases("3자간 등기명의신탁 수탁자 임의처분 이행불능 부당이득반환")
2. search_cases("3자간 등기명의신탁 신탁자 수탁자 직접 소유권이전등기 실체관계 부합")
3. search_cases("계약명의신탁 매도인 선의 대물변제 제3자 등기")
4. search_cases("계약명의신탁 계약체결 후 매도인 약의")
5. search_cases("계약명의신탁 매도인 약의 신탁자 소유권이전등기청구")

Claude Sonnet 4.5 (Max-Think) – 8 calls, **Correct** (①)

1. search_statutes("부동산 실권리자명의 등기에 관한 법률")
2. search_cases("3자간 명의신탁 부당이득반환")
3. search_cases("계약명의신탁 매도인 알지 못한 경우")
4. search_cases("계약명의신탁 명의수탁자 매수자금 반환 제3자 소유권이전등기")
5. get_case_content("2018다284233")
6. get_case_content("2017다257715")
7. search_cases("계약명의신탁 매도인 알고 있는 경우 명의신탁자 소유권이전등기청구")
8. get_statute_content("1178", "4")

Gemini 2.5 Pro (High) – 5 calls, **Incorrect** (③)